

Credit Card Approval Prediction using Machine Learning

Project Description:

Credit Card Approval Prediction is a machine learning-based web application that automates the credit card approval process by predicting whether an applicant is eligible for a credit card based on financial and demographic information. The system analyzes applicant details such as income, employment status, credit history, family size, and other relevant features to provide an instant approval or rejection prediction.

The project compares four classification algorithms—**Logistic Regression, Decision Tree, Random Forest, and XGBoost**—to identify the best-performing model. The selected model is integrated into a **Flask web application** for real-time prediction and deployed using **IBM Watson Machine Learning** for cloud-based access. The system reduces manual verification time, improves decision accuracy, and enhances customer experience.

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Scenarios

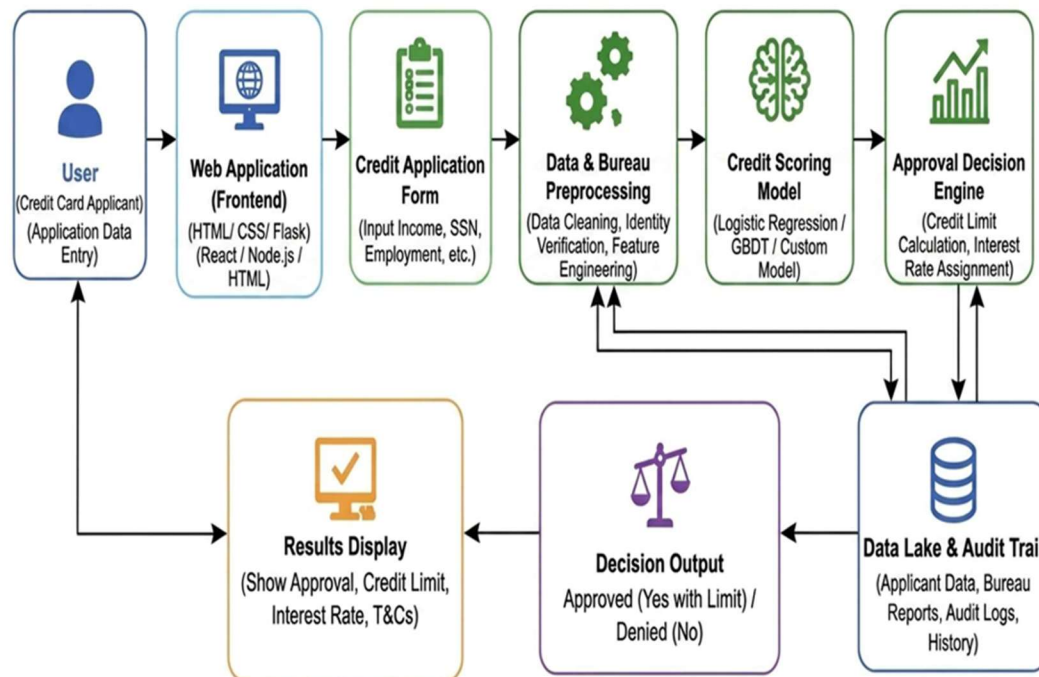
Scenario 1: A bank employee enters an applicant's financial and personal details into the web application. The trained machine learning model instantly predicts whether the application is likely to be approved or rejected, helping the bank process applications more efficiently.

Scenario 2: A financial institution screens applicants with poor credit history or previous loan defaults. The system identifies high-risk applicants using historical data and machine learning models, reducing financial risk.

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Scenario 3: A customer checks their eligibility before submitting a credit card application. The application predicts the likelihood of approval, helping customers avoid unnecessary rejections.

Technical Architecture:



Description:

The system follows a three-layer architecture consisting of a Frontend, Flask Backend, and Machine Learning Prediction Engine.

- The Frontend provides a simple interface for entering applicant details.
- The Flask Backend validates user input, preprocesses applicant data, and communicates with the trained machine learning model.
- The Machine Learning Engine predicts whether the application will be approved or rejected using the trained model.
- The prediction result is displayed instantly through the web interface and can also be accessed through IBM Watson Machine Learning cloud deployment.

Pre-requisites:

Python Programming

Vs code

Flask Framework

HTML, CSS, Bootstrap

Pandas

NumPy

Scikit-learn

XGBoost

IBM Watson Machine Learning

Project Workflow:

Milestone 1: Data Collection & Preprocessing

Collect Credit Card Approval Dataset
Handle Missing Values
Encode Categorical Features
Feature Selection
Split Dataset into Training and Testing

Milestone 2 : Machine Learning Model Development

Train Logistic Regression
Train Decision Tree
Train Random Forest
Train XGBoost
Compare Accuracy
Select Best Model

Milestone 3: Flask Application Development

Create Home Page
Develop Prediction Form
Load Trained Model (.pkl)
Predict Credit Card Approval
Display Prediction Result

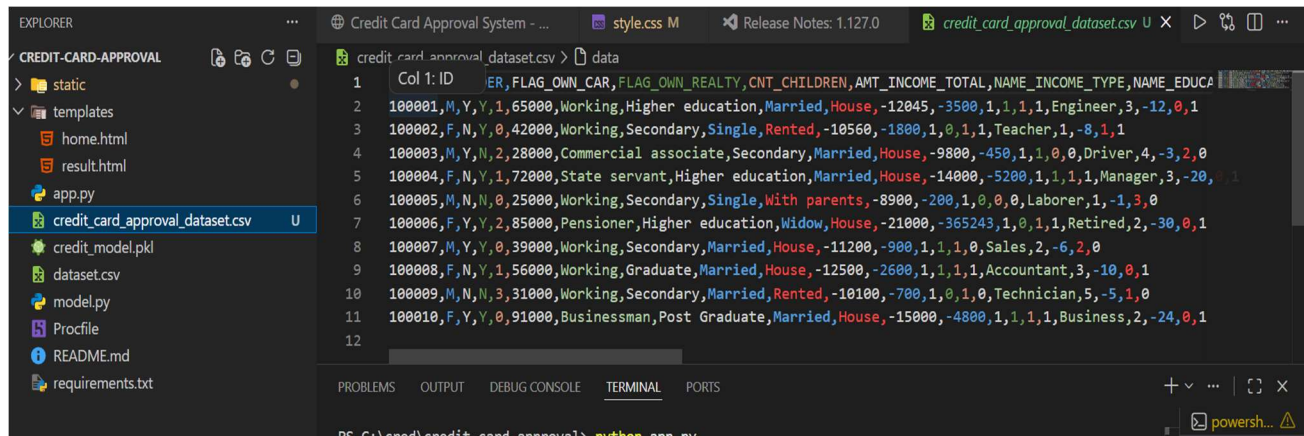
Milestone 4: Frontend Development

- Responsive User Interface
- Applicant Information Form
- Result Display Page
- Input Validation
- User-Friendly Design

Milestone 5: Conclusion

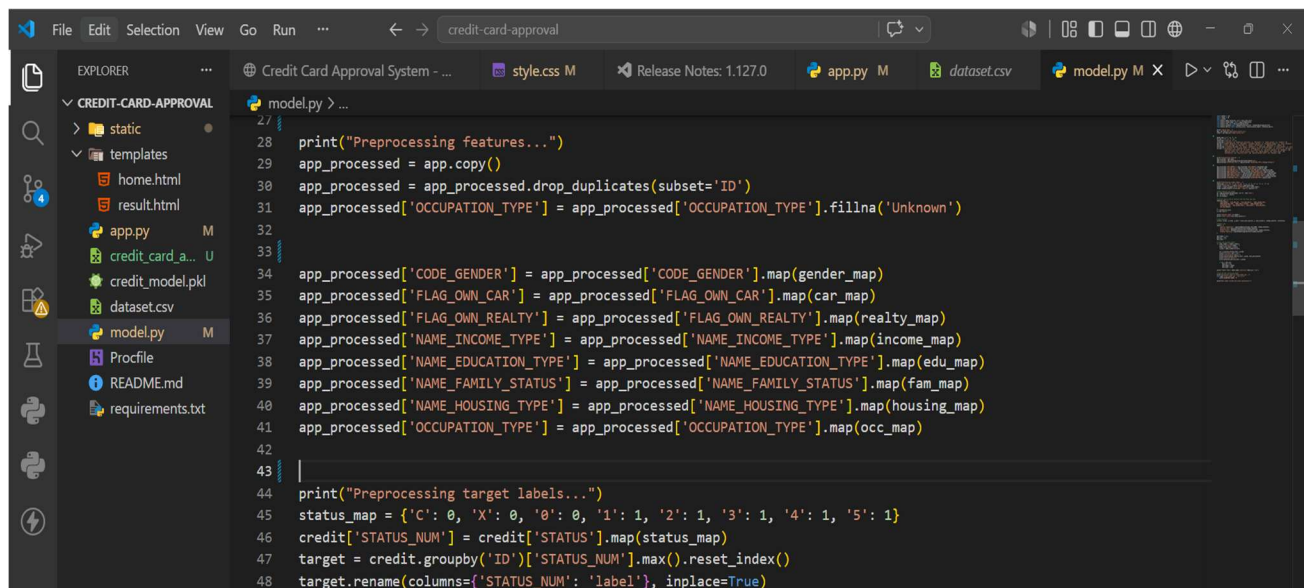
Evaluate Model Performance
Compare Machine Learning Algorithms
Document Results
Complete Project Deployment

Milestone 1: Data Collection & Preprocessing



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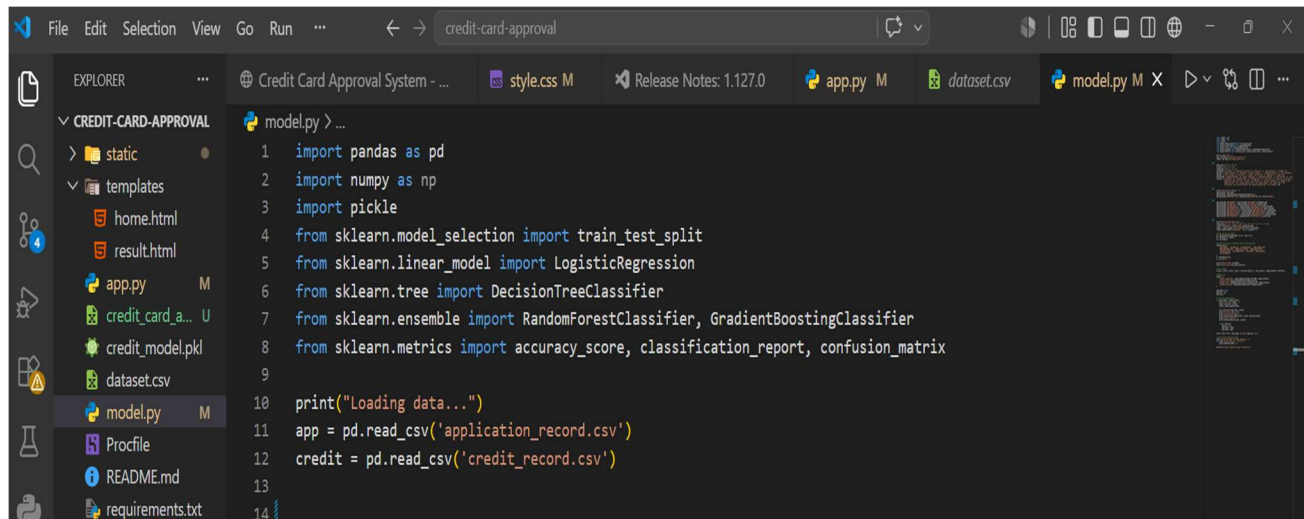
1  Col 1: ID      SEX, FLAG_OWN_CAR, FLAG_OWN_REALTY, CNT_CHILDREN, AMT_INCOME_TOTAL, NAME_INCOME_TYPE, NAME_EDUCA
2  100001,M,Y,Y,1,65000,Working,Higher education,Married,House,-12045,-3500,1,1,1,1,Engineer,3,-12,0,1
3  100002,F,N,Y,0,42000,Working,Secondary,Single,Rented,-10560,-1800,1,0,1,1,Teacher,1,-8,1,1
4  100003,M,Y,N,2,28000,Commercial associate,Secondary,Married,House,-9800,-450,1,1,0,0,Driver,4,-3,2,0
5  100004,F,N,Y,1,72000,State servant,Higher education,Married,House,-14000,-5200,1,1,1,1,Manager,3,-20,
6  100005,M,N,N,0,25000,Working,Secondary,Single,With parents,-8900,-200,1,0,0,0,Laborer,1,-1,3,0
7  100006,F,Y,Y,2,85000,Pensioner,Higher education,Widow,House,-21000,-365243,1,0,1,1,Retired,2,-30,0,1
8  100007,M,Y,Y,0,39000,Working,Secondary,Married,House,-11200,-900,1,1,1,0,Sales,2,-6,2,0
9  100008,F,N,Y,1,56000,Working,Graduate,Married,House,-12500,-2600,1,1,1,1,Accountant,3,-10,0,1
10 100009,M,N,N,3,31000,Working,Secondary,Married,Rented,-10100,-700,1,0,1,0,Technician,5,-5,1,0
11 100010,F,Y,Y,0,91000,Businessman,Post Graduate,Married,House,-15000,-4800,1,1,1,1,Business,2,-24,0,1
12
  
```



```

27
28 print("Preprocessing features...")
29 app_processed = app.copy()
30 app_processed = app_processed.drop_duplicates(subset='ID')
31 app_processed['OCCUPATION_TYPE'] = app_processed['OCCUPATION_TYPE'].fillna('Unknown')
32
33
34 app_processed['CODE_GENDER'] = app_processed['CODE_GENDER'].map(gender_map)
35 app_processed['FLAG_OWN_CAR'] = app_processed['FLAG_OWN_CAR'].map(car_map)
36 app_processed['FLAG_OWN_REALTY'] = app_processed['FLAG_OWN_REALTY'].map(realty_map)
37 app_processed['NAME_INCOME_TYPE'] = app_processed['NAME_INCOME_TYPE'].map(income_map)
38 app_processed['NAME_EDUCATION_TYPE'] = app_processed['NAME_EDUCATION_TYPE'].map(edu_map)
39 app_processed['NAME_FAMILY_STATUS'] = app_processed['NAME_FAMILY_STATUS'].map(fam_map)
40 app_processed['NAME_HOUSING_TYPE'] = app_processed['NAME_HOUSING_TYPE'].map(housing_map)
41 app_processed['OCCUPATION_TYPE'] = app_processed['OCCUPATION_TYPE'].map(occ_map)
42
43
44 print("Preprocessing target labels...")
45 status_map = {'C': 0, 'X': 0, '0': 0, '1': 1, '2': 1, '3': 1, '4': 1, '5': 1}
46 credit['STATUS_NUM'] = credit['STATUS'].map(status_map)
47 target = credit.groupby('ID')['STATUS_NUM'].max().reset_index()
48 target.rename(columns={'STATUS_NUM': 'label'}, inplace=True)
  
```

Milestone 2 : Machine Learning Model Development

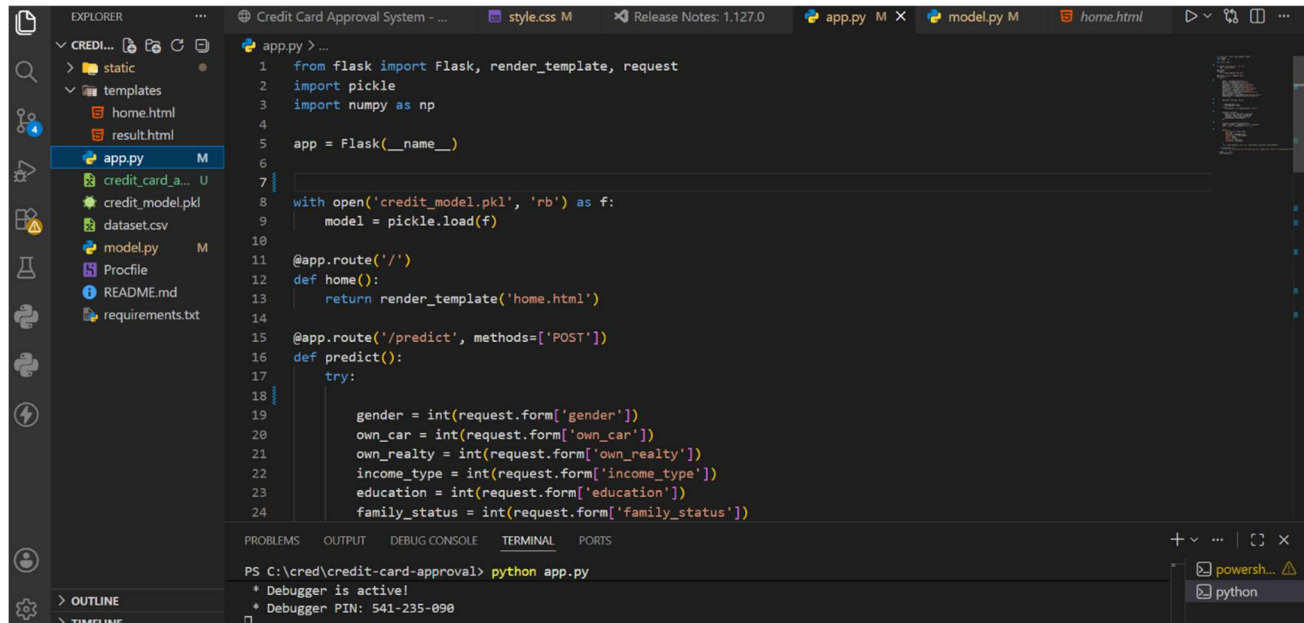


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1  import pandas as pd
2  import numpy as np
3  import pickle
4  from sklearn.model_selection import train_test_split
5  from sklearn.linear_model import LogisticRegression
6  from sklearn.tree import DecisionTreeClassifier
7  from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
8  from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
9
10 print("Loading data...")
11 app = pd.read_csv('application_record.csv')
12 credit = pd.read_csv('credit_record.csv')
13
14

```

Milestone 3: Flask Application Development



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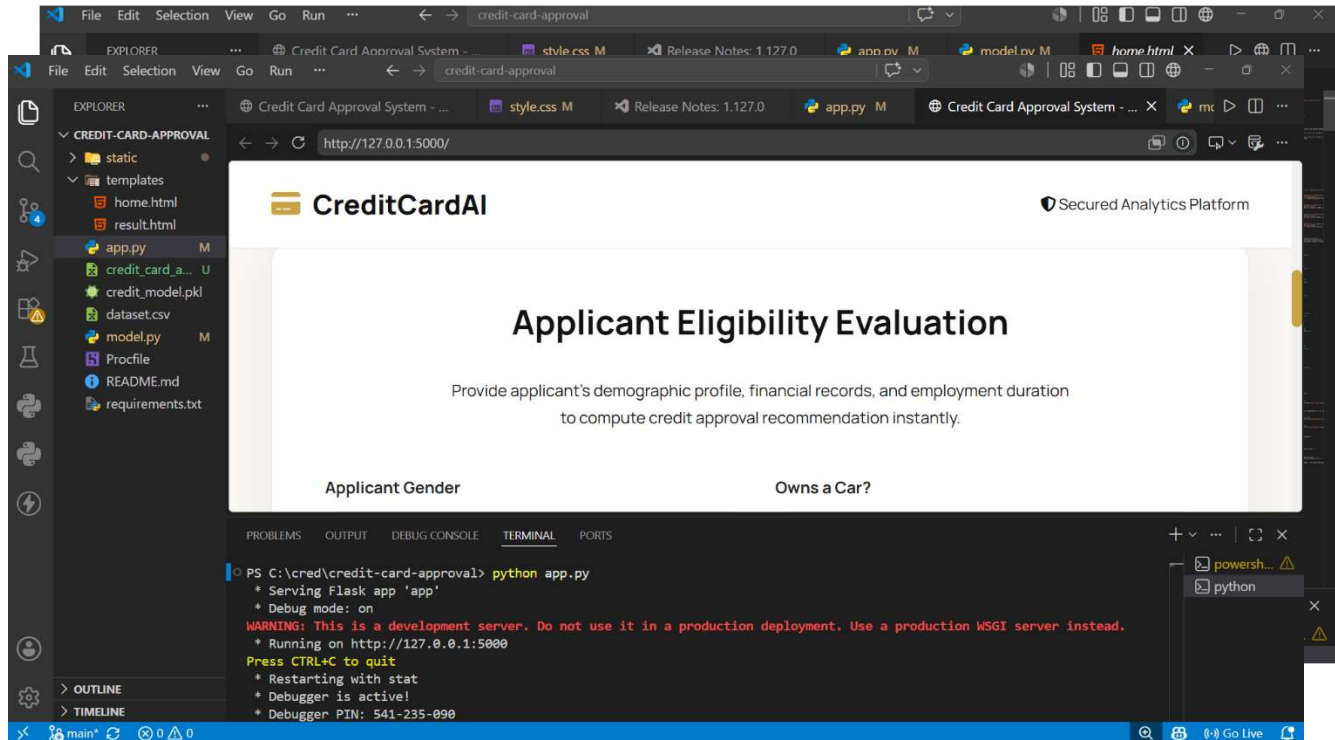
1  from flask import Flask, render_template, request
2  import pickle
3  import numpy as np
4
5  app = Flask(__name__)
6
7
8  with open('credit_model.pkl', 'rb') as f:
9      model = pickle.load(f)
10
11 @app.route('/')
12 def home():
13     return render_template('home.html')
14
15 @app.route('/predict', methods=['POST'])
16 def predict():
17     try:
18
19         gender = int(request.form['gender'])
20         own_car = int(request.form['own_car'])
21         own_realty = int(request.form['own_realty'])
22         income_type = int(request.form['income_type'])
23         education = int(request.form['education'])
24         family_status = int(request.form['family_status'])

```

PS C:\cred\credit-card-approval> python app.py

* Debugger is active!
 * Debugger PIN: 541-235-090

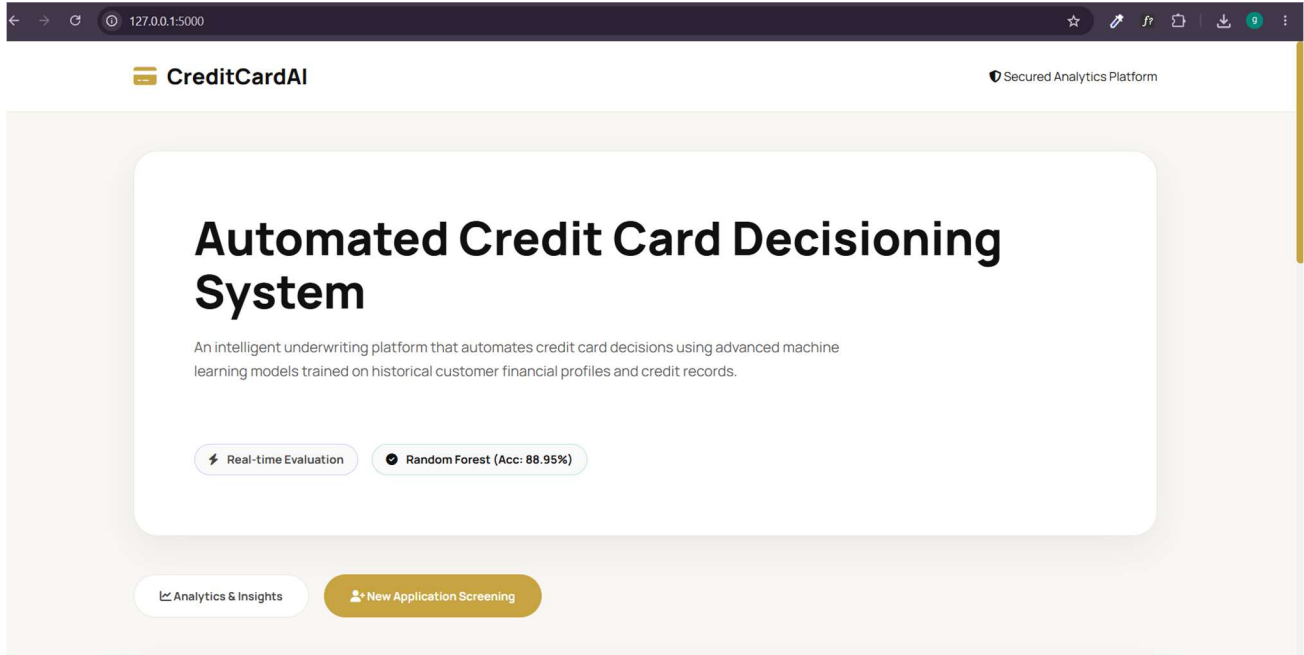
Milestone 4: Frontend Development



Milestone 5: Conclusion

Exploring the Website's Web Pages:

Home / Generator Page:

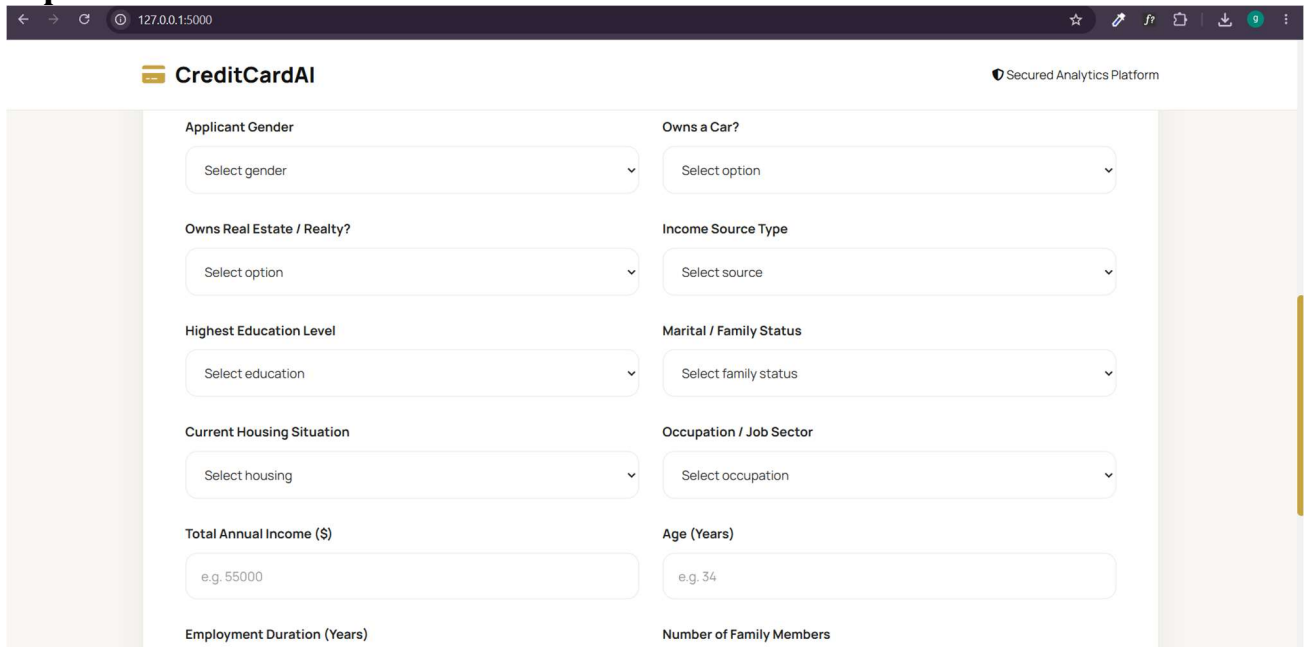


The screenshot shows the home page of the CreditCardAI website. The browser address bar displays "127.0.0.1:5000". The website header includes the "CreditCardAI" logo and a "Secured Analytics Platform" indicator. The main content area features a large white card with the title "Automated Credit Card Decisioning System". Below the title, a subtitle reads: "An intelligent underwriting platform that automates credit card decisions using advanced machine learning models trained on historical customer financial profiles and credit records." Two buttons are present: "Real-time Evaluation" and "Random Forest (Acc: 88.95%)". At the bottom of the card, there are two navigation buttons: "Analytics & Insights" and "New Application Screening".

Description:

Displays the project title and provides a form for entering applicant information required for credit card approval prediction.

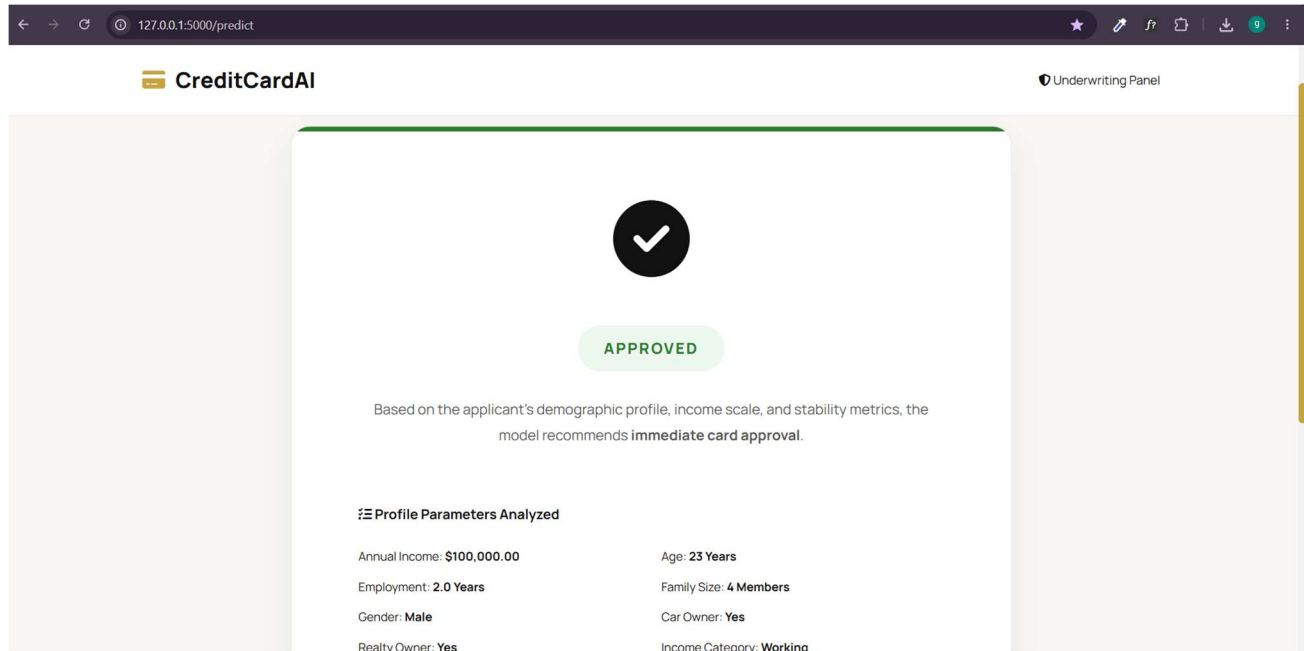
Input Section:



The screenshot shows the input section of the CreditCardAI website. The browser address bar displays "127.0.0.1:5000". The website header includes the "CreditCardAI" logo and a "Secured Analytics Platform" indicator. The input form is organized into two columns. The left column contains: "Applicant Gender" (dropdown), "Owns Real Estate / Realty?" (dropdown), "Highest Education Level" (dropdown), "Current Housing Situation" (dropdown), "Total Annual Income (\$)" (text input with placeholder "e.g. 55000"), and "Employment Duration (Years)". The right column contains: "Owns a Car?" (dropdown), "Income Source Type" (dropdown), "Marital / Family Status" (dropdown), "Occupation / Job Sector" (dropdown), "Age (Years)" (text input with placeholder "e.g. 34"), and "Number of Family Members".

Allows users to enter financial and demographic details such as income, employment status, family members, education level, housing type, and credit history.

Results Section:



Displays whether the credit card application is Approved or Rejected along with the prediction generated by the machine learning model.

Conclusion:

The Credit Card Approval Prediction system demonstrates how machine learning can automate the credit card approval process by providing fast, accurate, and reliable predictions. By comparing multiple classification algorithms and deploying the best-performing model through Flask and IBM Watson Machine Learning, the project minimizes manual effort, improves decision-making, and enhances customer satisfaction.